

## TAIL ROTOR FAILURE CONDITIONS

### 3.5 TAIL ROTOR FAILURE CONDITIONS

#### 3.5.1 Tail Rotor Drive Failure - Hover

##### Conditions/Indications

Complete loss of tail rotor thrust

- Rapid uncontrolled right yawing motion

##### Procedure

##### • HOVER IGE

- |                     |             |
|---------------------|-------------|
| 1. Collective lever | – Full down |
|---------------------|-------------|

and simultaneously

- |                     |             |
|---------------------|-------------|
| 2. Landing attitude | – Establish |
|---------------------|-------------|

After landing:

- |                                      |           |
|--------------------------------------|-----------|
| 3. Double engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

##### • HOVER OGE

- |                     |             |
|---------------------|-------------|
| 1. Collective lever | – Full down |
|---------------------|-------------|

If height permits:

- |             |        |
|-------------|--------|
| 2. Airspeed | – Gain |
|-------------|--------|

- |                                      |           |
|--------------------------------------|-----------|
| 3. Double engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

- |                     |                                             |
|---------------------|---------------------------------------------|
| 4. Collective lever | – Raise to stop descent and cushion landing |
|---------------------|---------------------------------------------|

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.2 Tail Rotor Drive Failure - Flight

#### Conditions/Indications

Complete loss of tail rotor thrust which may be caused by a severed drive shaft and/or separation of the tail rotor assembly from the helicopter.

- Rapid right yawing motion eventually combined with a right roll and downward pitch.

#### Procedure

**WARNING** EXTENDED FLIGHT IS NOT POSSIBLE AFTER TAIL ROTOR DRIVE SYSTEM FAILURE. AUTOROTATION MUST BE ENTERED IMMEDIATELY.

|                            |                                                                            |
|----------------------------|----------------------------------------------------------------------------|
| <b>1. Collective lever</b> | – <b>Full down immediately, maintain <math>N_{RO}</math> within limits</b> |
|----------------------------|----------------------------------------------------------------------------|

|                    |                              |
|--------------------|------------------------------|
| <b>2. Airspeed</b> | – <b>75 KIAS recommended</b> |
|--------------------|------------------------------|

**NOTE** Airspeed indication may be unreliable due to right yaw flight condition.

|                                             |                  |
|---------------------------------------------|------------------|
| <b>3. Double engine emergency shut-down</b> | – <b>Perform</b> |
|---------------------------------------------|------------------|

**WARNING** LANDING WITHOUT THE TAIL ROTOR OPERATING IS HAZARDOUS AS THE HELICOPTER WILL TOUCH DOWN IN AN EXTREME YAW ATTITUDE AND PROBABLY ROLL OVER.

|                               |                                                                 |
|-------------------------------|-----------------------------------------------------------------|
| <b>4. Modified side flare</b> | – <b>As necessary to obtain zero groundspeed upon touchdown</b> |
|-------------------------------|-----------------------------------------------------------------|

**NOTE** The rate of descent will be higher than normal due to the extreme yaw attitude and will require a modified side flare to reduce the groundspeed to as near zero as possible to prevent the helicopter from rolling over upon touchdown.

|                            |                                                    |
|----------------------------|----------------------------------------------------|
| <b>5. Collective lever</b> | – <b>Raise to stop descent and cushion landing</b> |
|----------------------------|----------------------------------------------------|

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.3 Fixed-pitch Tail Rotor Control Failures

#### Conditions/Indications

Fixed-pitch tail rotor failure

- No directional response after pedal movement
- Locked pedals

#### Procedure

The emergency procedures for a tail rotor control failure will vary depending on flight conditions, power setting and mass of the helicopter.

Lowering or raising the collective lever will result in a left or right yawing motion to a degree dependent upon the tail rotor thrust remaining. Additionally, at high airspeeds, an ever increasing yaw oscillation may occur, however, is dampened by reducing airspeed to approx 60 KIAS and, if possible, increasing the power setting.

Approach and landing should be made with a suitable rate of descent, controlling sideslip angle with the collective lever. If the situation permits, a crosswind landing should be performed to minimize the amount of sideslip. Before touchdown, the ground speed should be reduced to a minimum. Touchdown so that the skids are aligned with the direction of landing.

#### Right Pedal Fixed-pitch Settings

Raising collective lever will result in a right yawing motion. Lower collective lever to control sideslip and touch down with forward speed.

#### Left Pedal Fixed-pitch Settings

Lowering collective lever will result in a left yawing motion. Approach with a right sideslip and reduce ground speed to allow for raising the collective lever as necessary to touchdown.

## TAIL ROTOR FAILURE CONDITIONS

### 3.5.4 Pedal Vibrations

#### Conditions/Indications

Impending tail rotor system failure

- Unusual pedal vibrations

#### Procedure

**LAND AS SOON AS POSSIBLE**